

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/Correction_X/G_DEB_SAALKe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 3303, C2 = 3666, C3 = 2212)

text_priors <- '
Distrib (Fm,          TruncNormal,          0.1,      0.2, 0.0001,      0.8);
Distrib (pAm,         TruncNormal_cv,        100,      0.2, 40,      2000);
Distrib (r_pAm_pM,    TruncNormal_cv,        0.8,      0.1, 0.1,      2);
Distrib (v,           TruncNormal_cv,        0.18505,   0.1, 0.001,    2);
Distrib (kap,         Uniform,                0,        1);
Distrib (kapA,        Uniform,                0.1,      0.9);
Distrib (Ehp,         TruncNormal_cv,        100,      0.2, 40,      300);
Distrib (E_coc,       TruncNormal_cv,        30,      0.2, 1,      300);
Distrib (rOM_ClxHorse, LogUniform,          0.00001,   0.3);
Distrib (K,           Uniform,                0.00001,   2);
Distrib (dv,          TruncNormal_cv,        2,      0.5, 0,      10);
Distrib (wE,          LogUniform,          0.000000001, 0.1);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

# f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter, RTOL, ATOL)

# ./mcsim.DEBcali DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

Fm.1. pAm.1. r_pAm_pM.1. v.1. kap.1. kapA.1.

Result_mode	0.006512680	164.13300	0.76851600	0.19488700	0.93167800	0.52960800
2.5%	0.001499760	148.83600	0.65120400	0.15767500	0.87750500	0.44435600
97.5%	0.010023400	195.48130	0.89848460	0.22356100	0.96014000	0.60678900
Min	0.001118520	104.83800	0.58454400	0.13210200	0.12674200	0.21707400
Max	0.320981000	208.16100	0.98192400	0.25220000	0.96710600	0.66045000
Result_mean	0.005359986	170.35497	0.76739107	0.19103584	0.92219208	0.52472837
Result_sd	0.003614076	12.76465	0.06051947	0.01710499	0.02182621	0.04283091
	Ehp.1.	E_coc.1.	rOM_ClxHorse.1.	K.1.	dv.1.	
Result_mode	90.55620	27.346300	0.007890700	0.04272900	4.0759400	
2.5%	52.94129	21.765500	0.004672943	0.00821440	3.0602200	
97.5%	132.31610	35.735200	0.010120045	0.06306139	5.6792825	
Min	43.52050	17.504600	0.001112680	0.00548910	0.5220400	
Max	163.89000	41.906000	0.289479000	1.59821000	6.6237000	
Result_mean	96.87672	27.680472	0.007335517	0.03308086	4.3224676	
Result_sd	19.72693	3.631482	0.002179768	0.01884351	0.6551622	
	wE.1.	Sigma_W.1.				
Result_mode	2.592620e-09	0.11884400				
2.5%	1.452420e-09	0.10843000				
97.5%	1.077080e-05	0.13727500				
Min	1.041680e-09	0.09458450				
Max	1.318410e-05	5.52230000				
Result_mean	3.332705e-06	0.12241158				
Result_sd	4.005932e-06	0.05252306				

Number of observations, n = 277
 Number of parameters, k = 12
 Log-likelihood, LL = -18.03382
 BIC = 103.5559

Potential scale reduction factors:

	Point est.	Upper C.I.
dv.1.	1.33	1.89
E_coc.1.	1.14	1.45
Ehp.1.	2.35	4.07
Fm.1.	2.32	4.68
K.1.	2.12	4.10
kap.1.	2.47	4.74
kapA.1.	1.43	2.09
pAm.1.	1.06	1.18
r_pAm_pM.1.	1.03	1.08

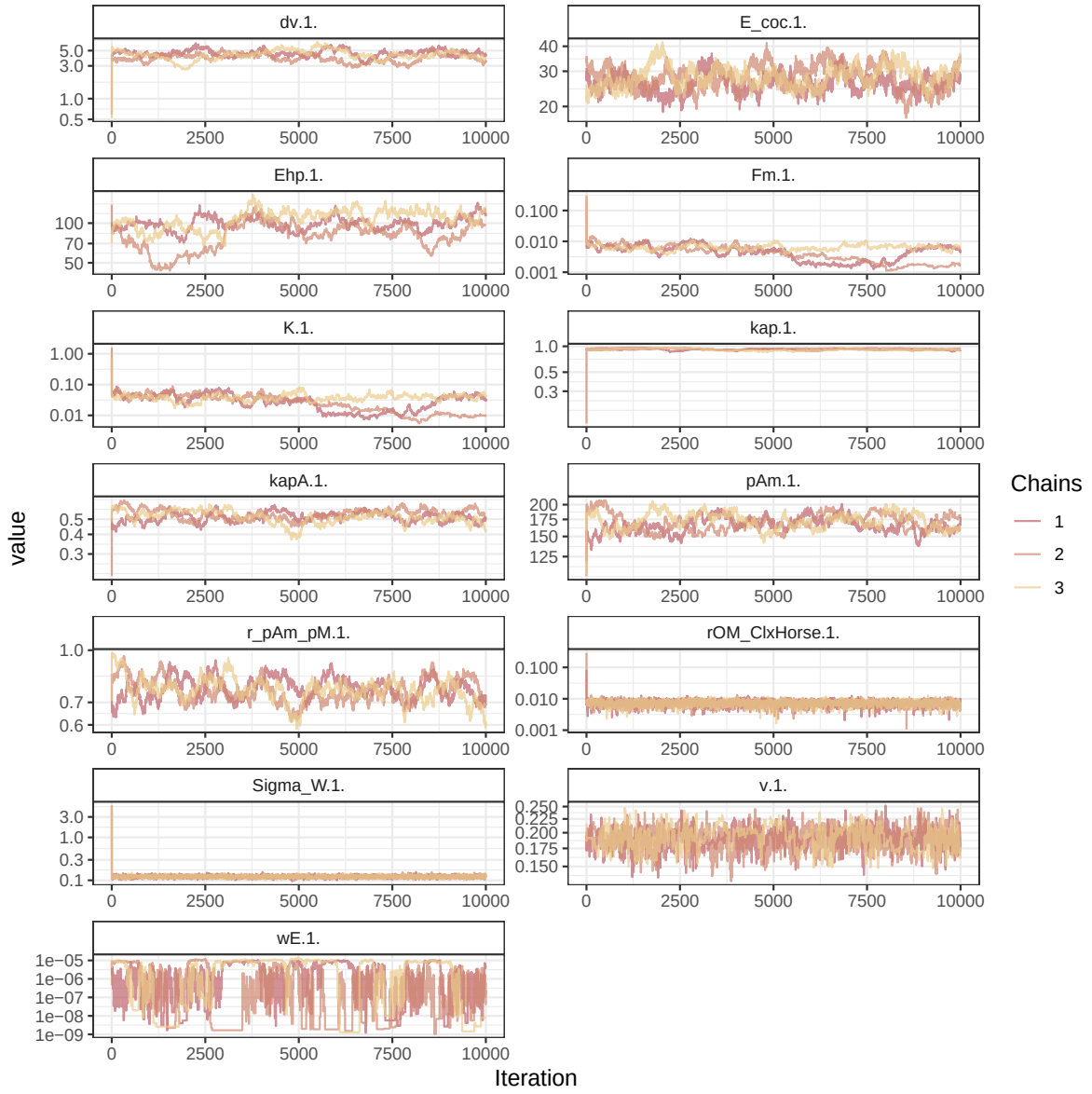


Figure 1: Parameters chains. Only the first iteration and the last `Nb_iter_kept` are represented.

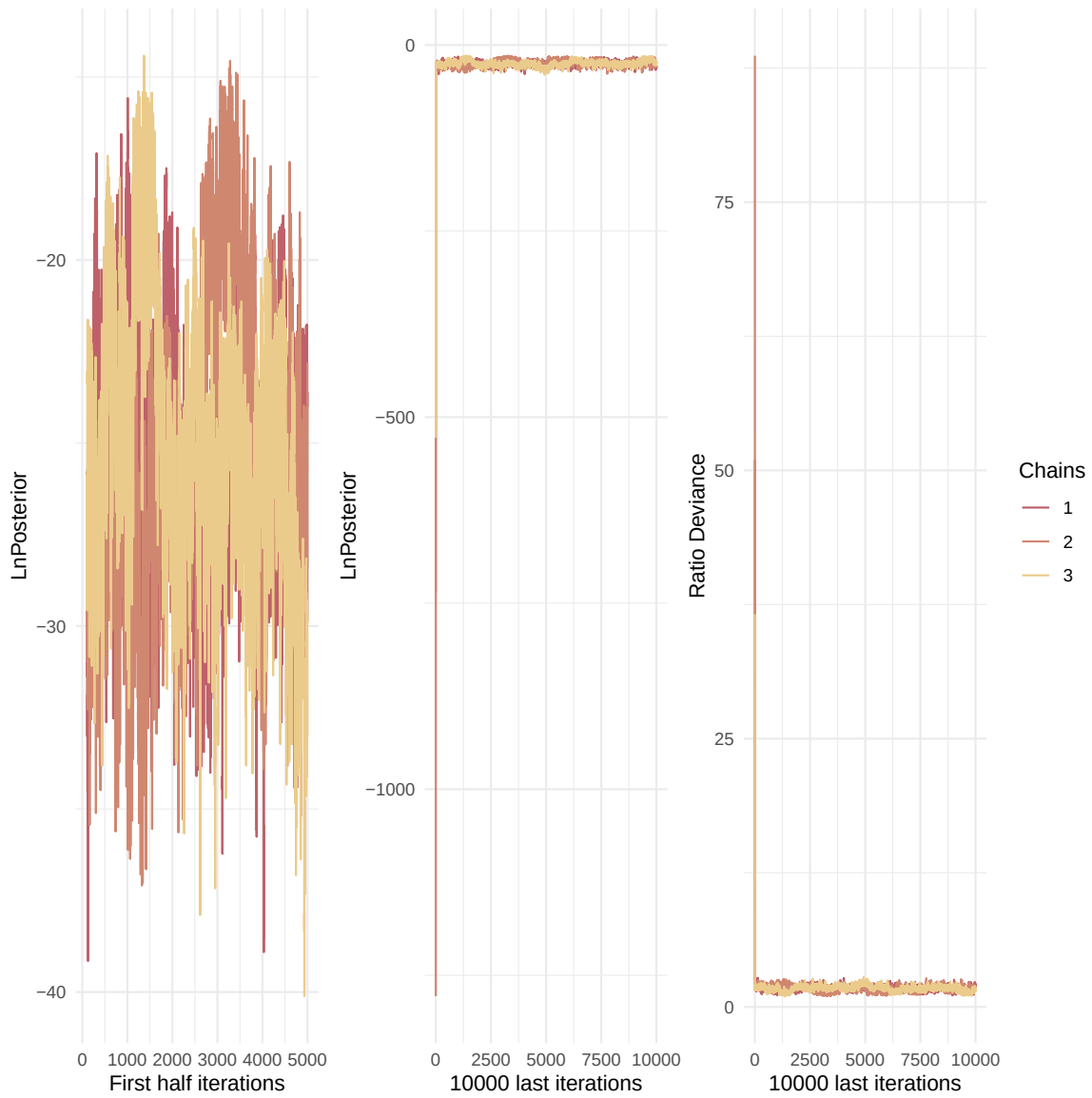


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

rOM_ClxHorse.1.	1.00	1.01
Sigma_W.1.	1.00	1.01
v.1.	1.01	1.02
wE.1.	1.36	2.24

Multivariate psrf

2.62

```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

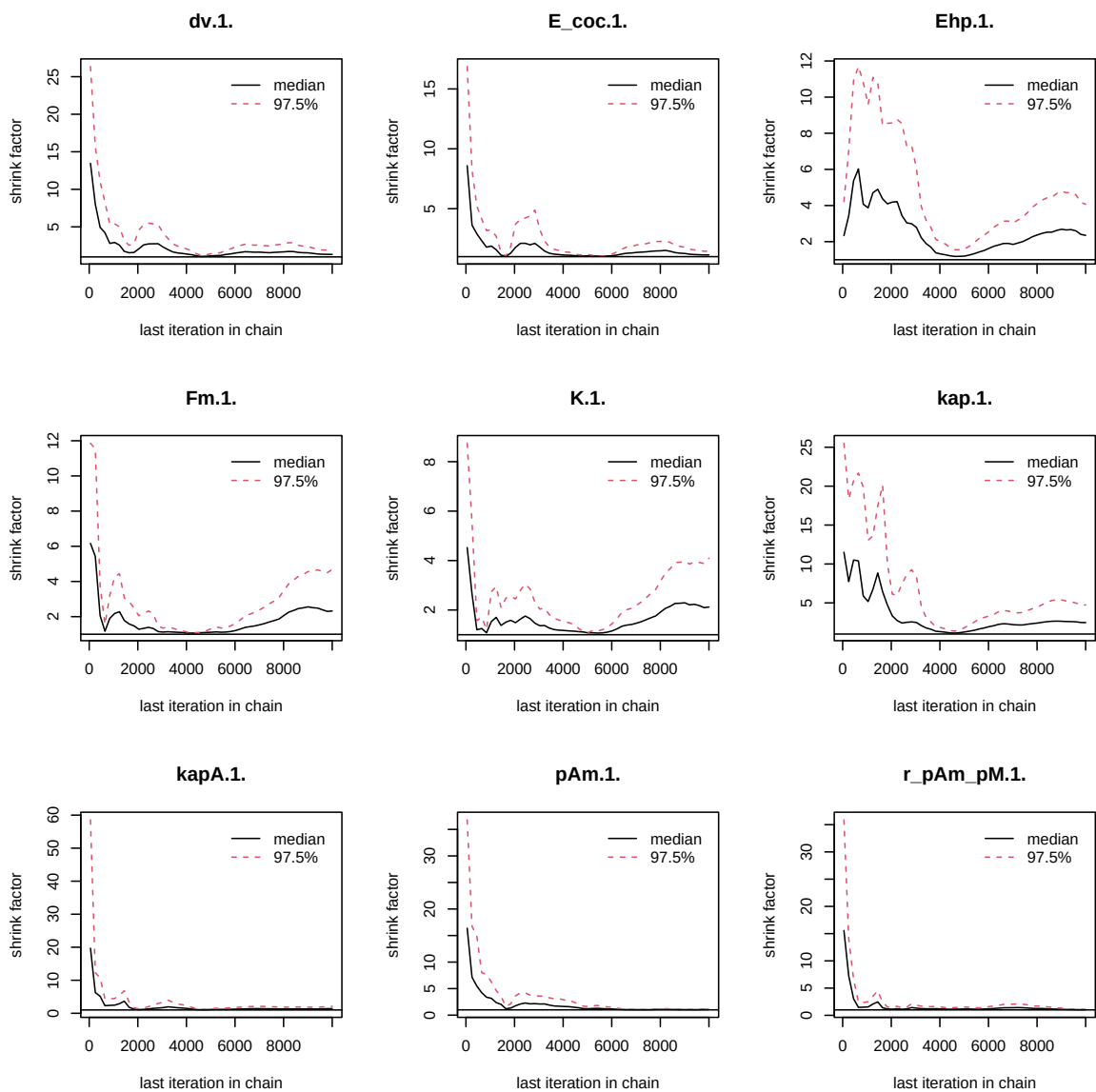


Figure 3: Chains convergence.

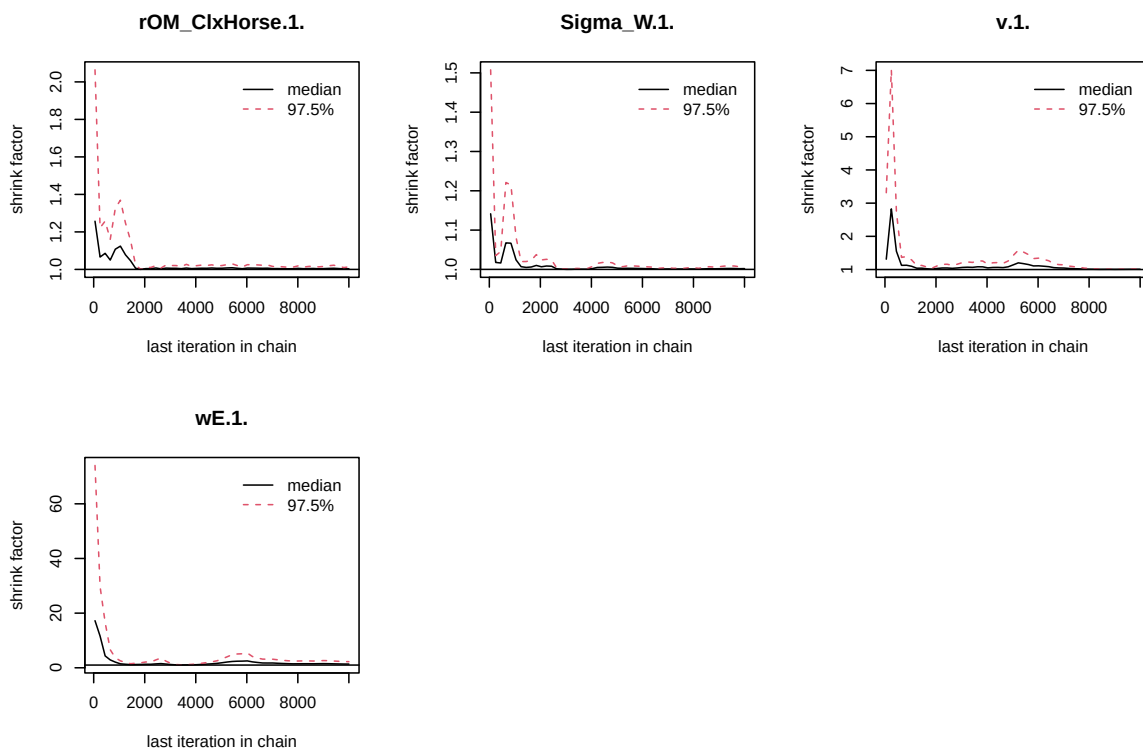


Figure 4: Chains convergence.

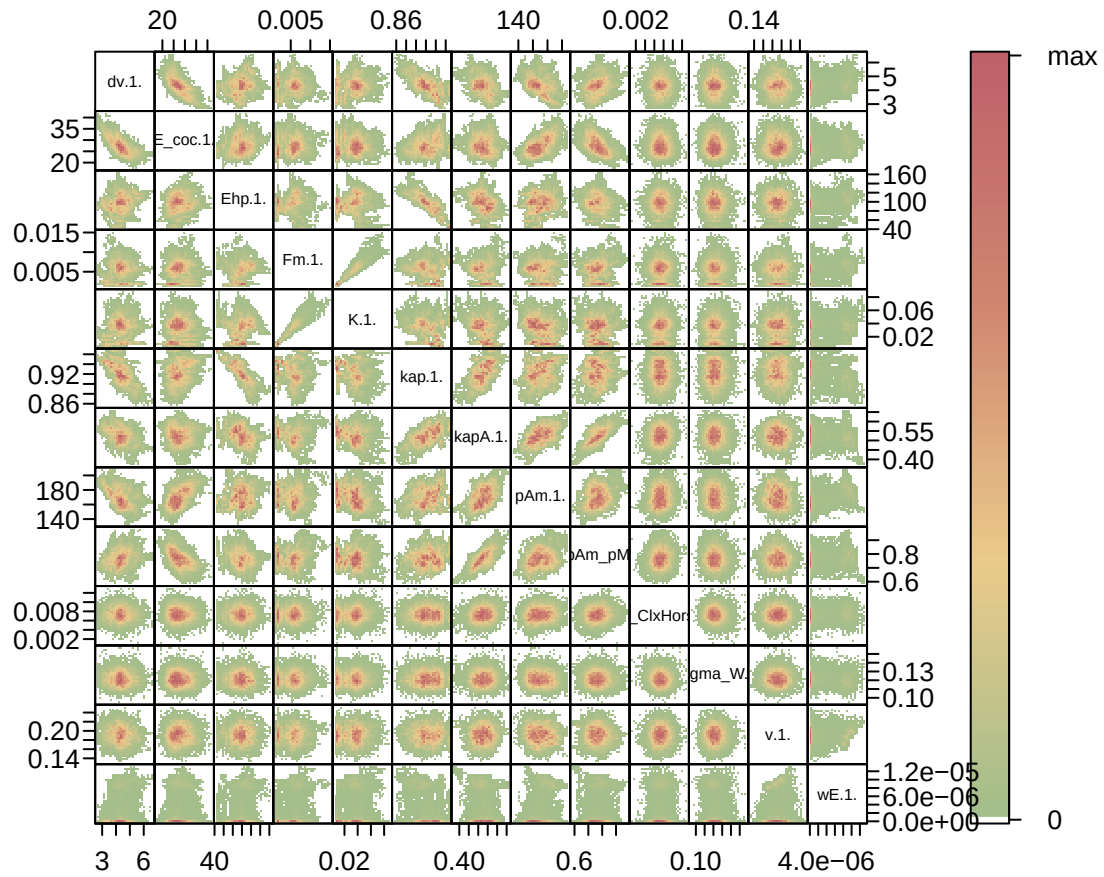


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

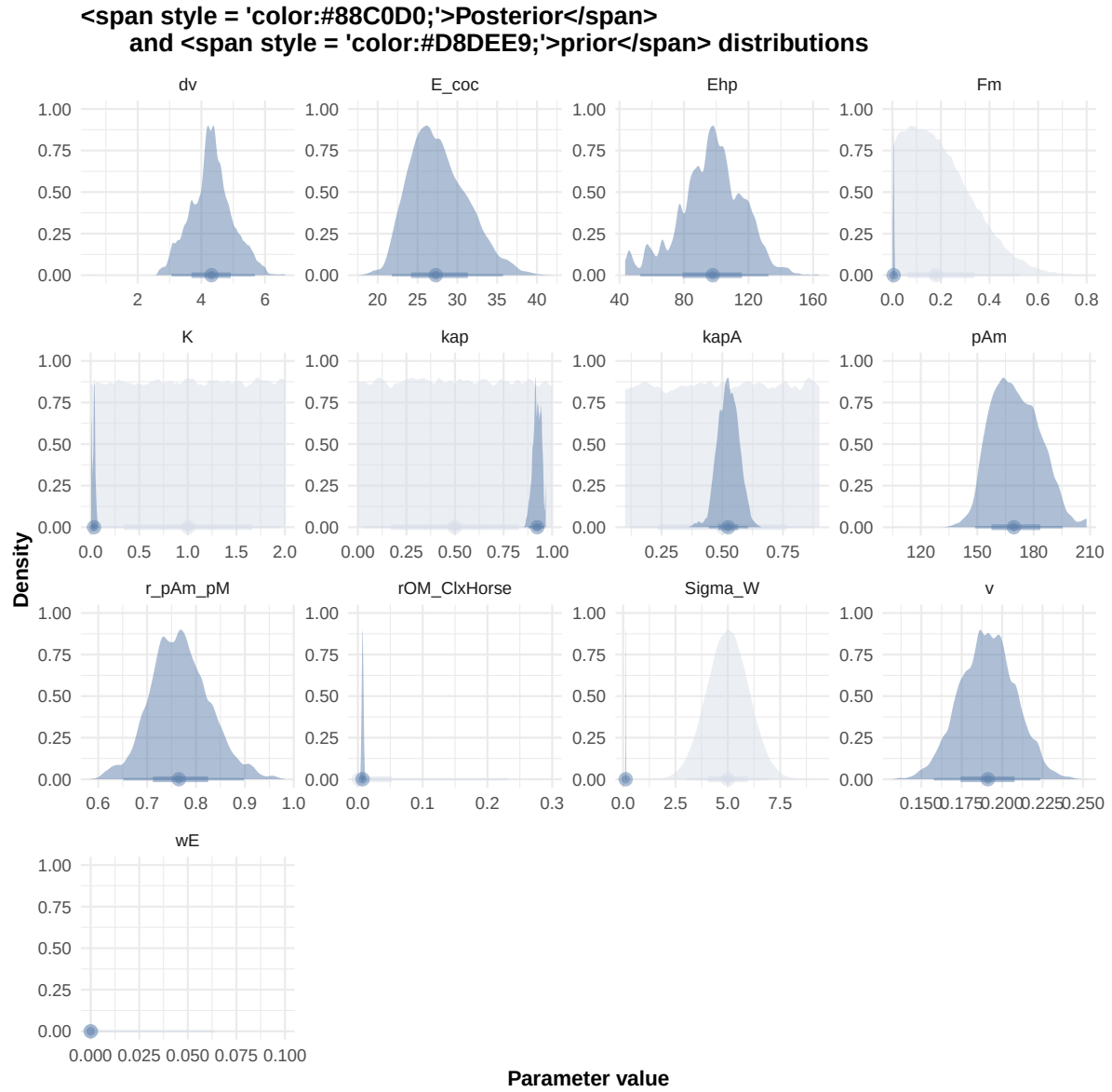


Figure 6: Distributions of the priors and the posteriors.


```
#####
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
)
```

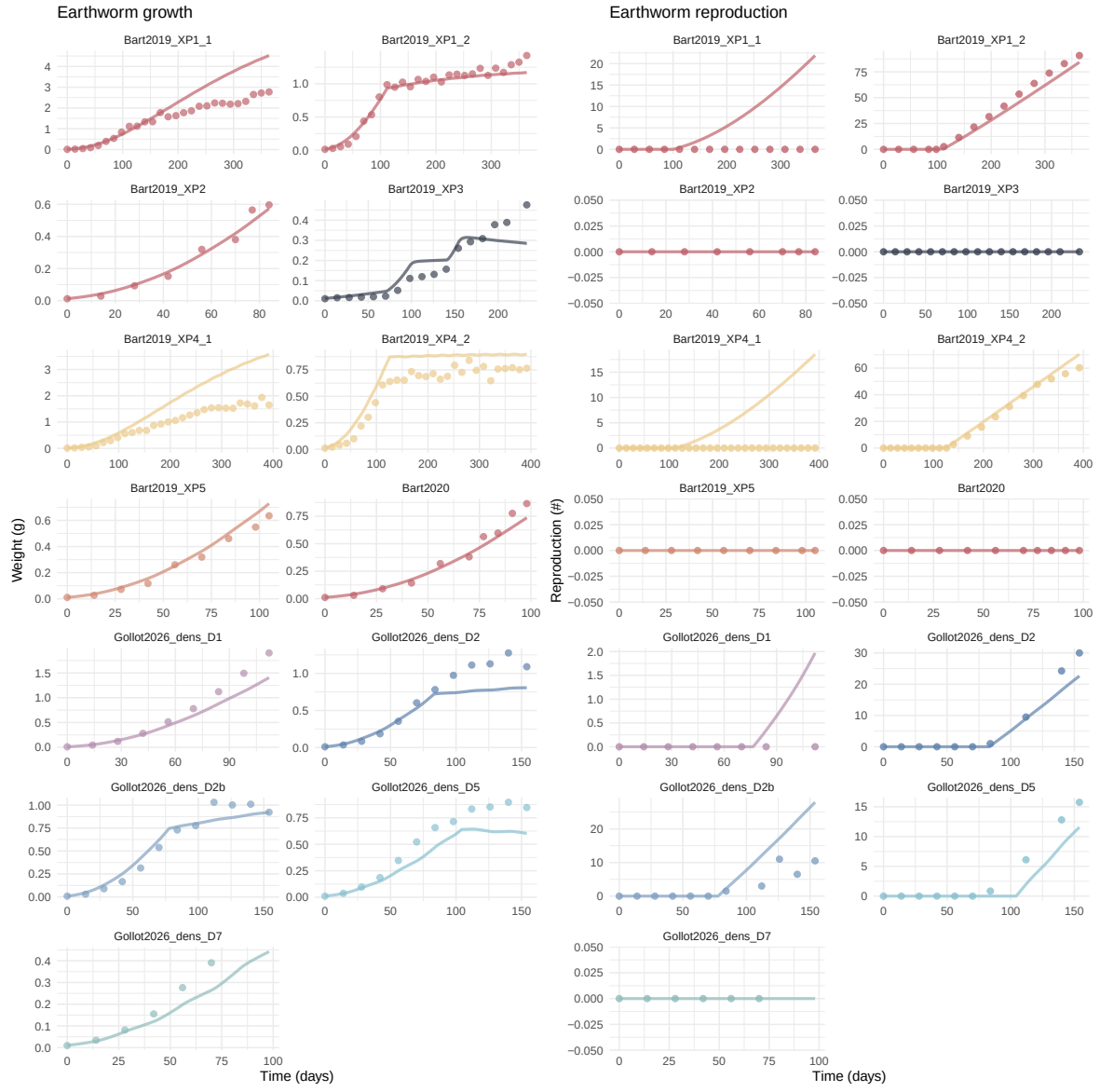
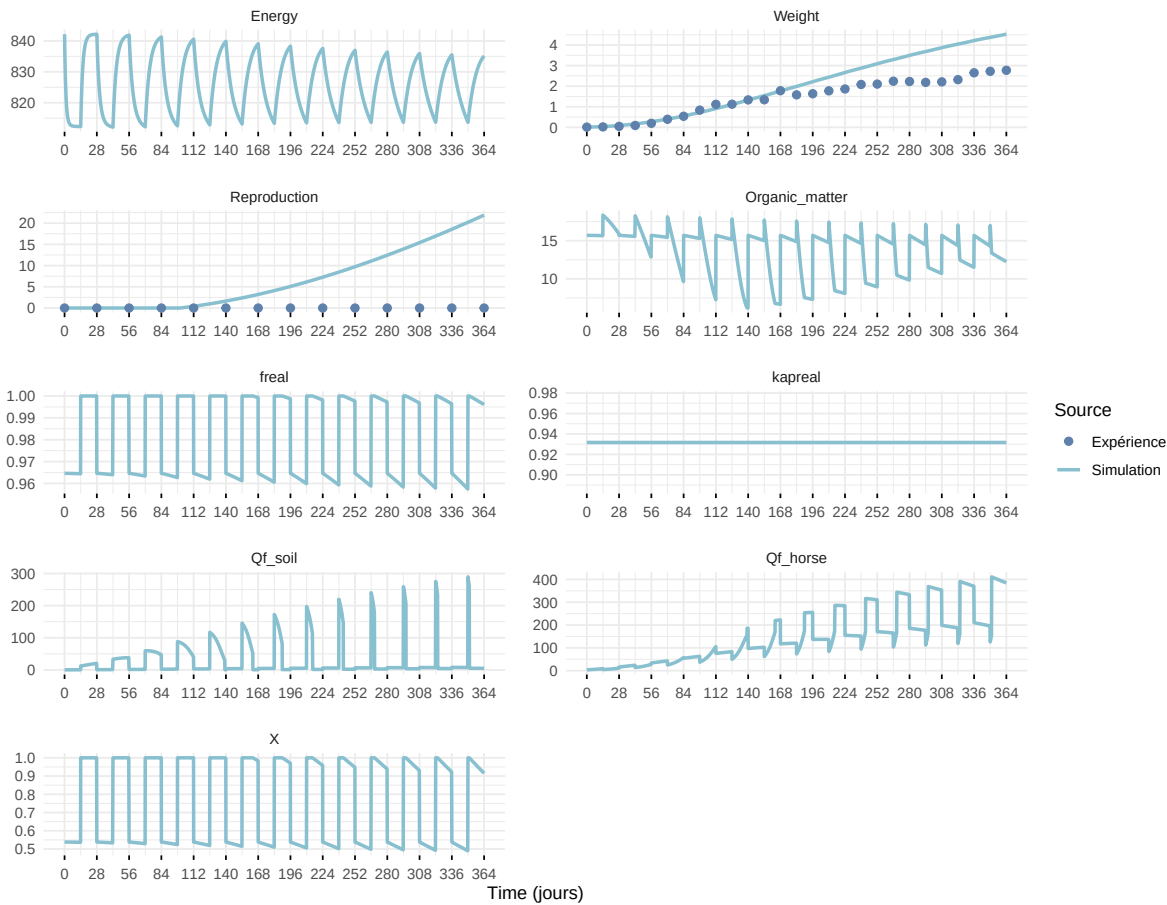
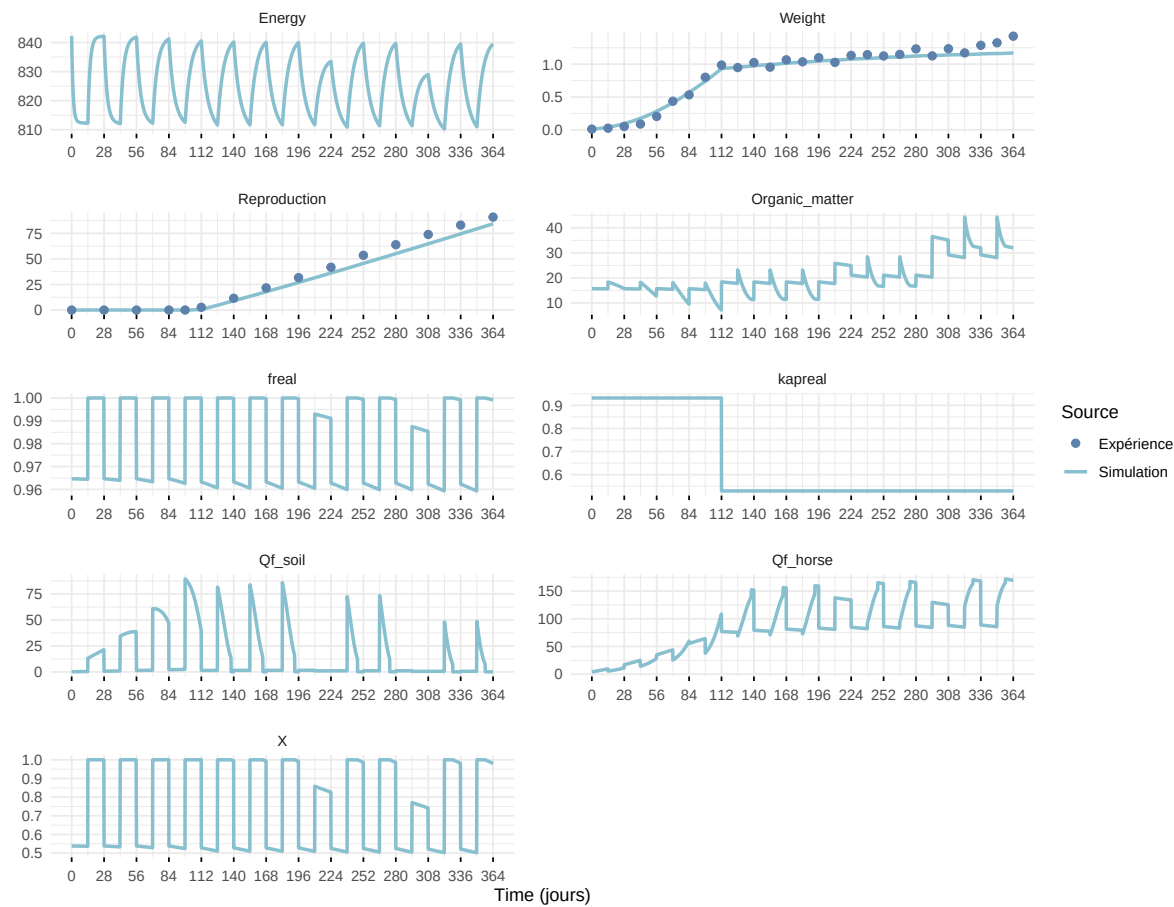


Figure 7: Simulations Weight and Reproduction.

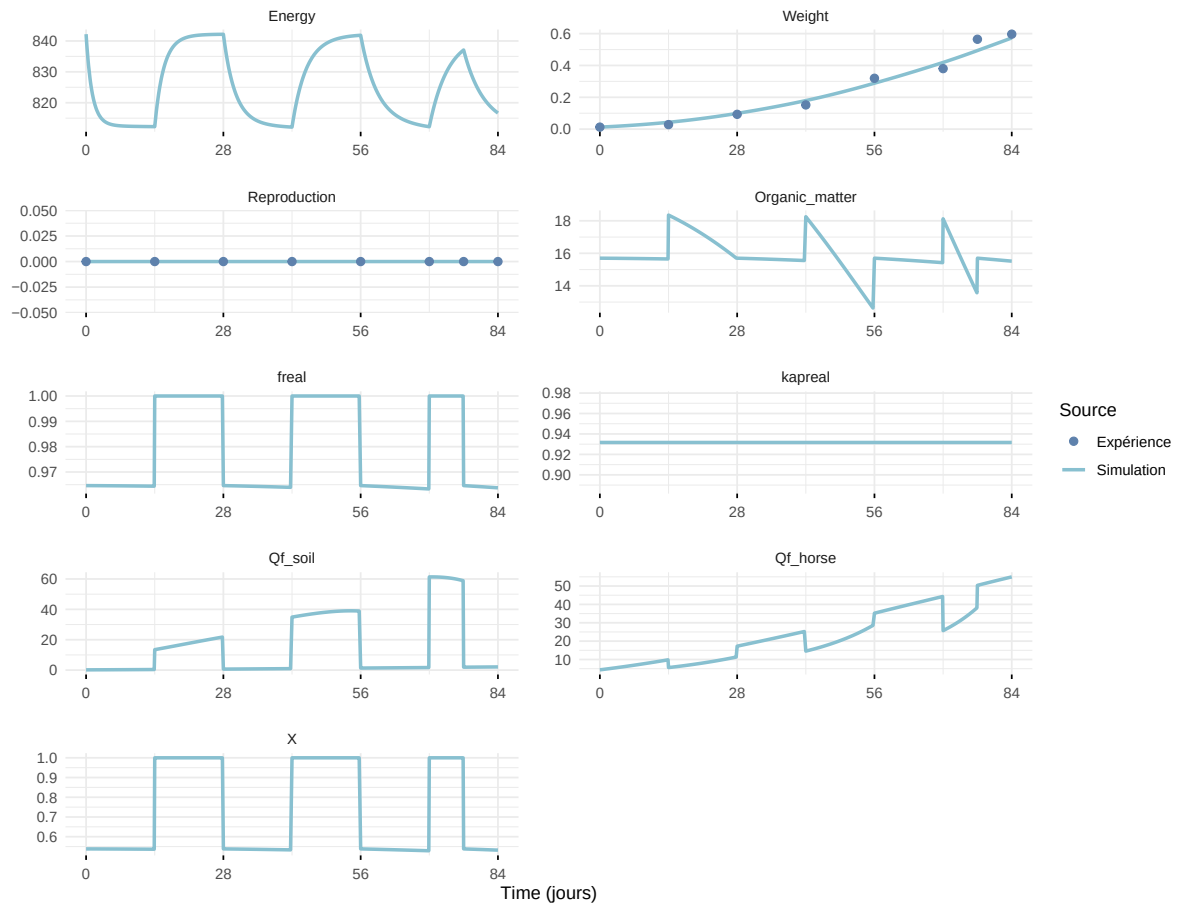
Expérience : Bart2019_XP1_1



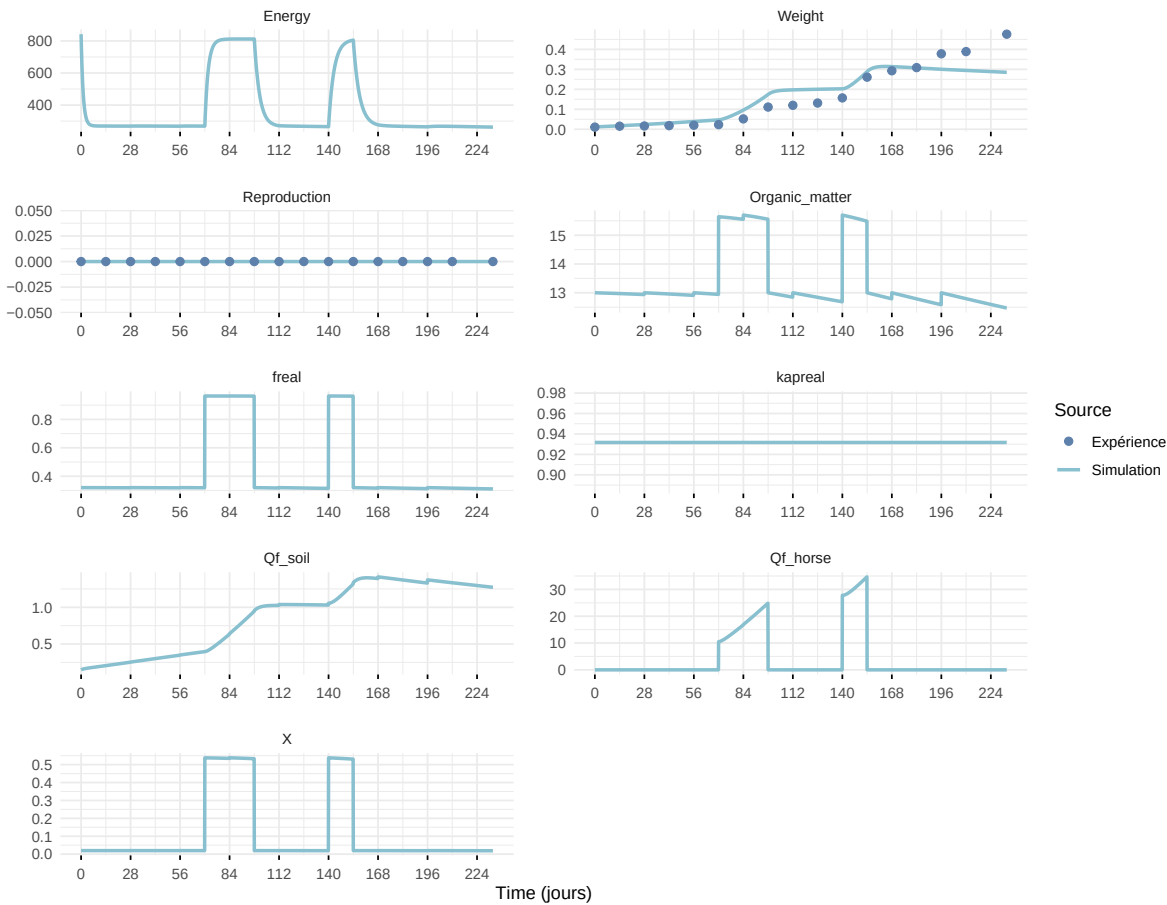
Expérience : Bart2019_XP1_2



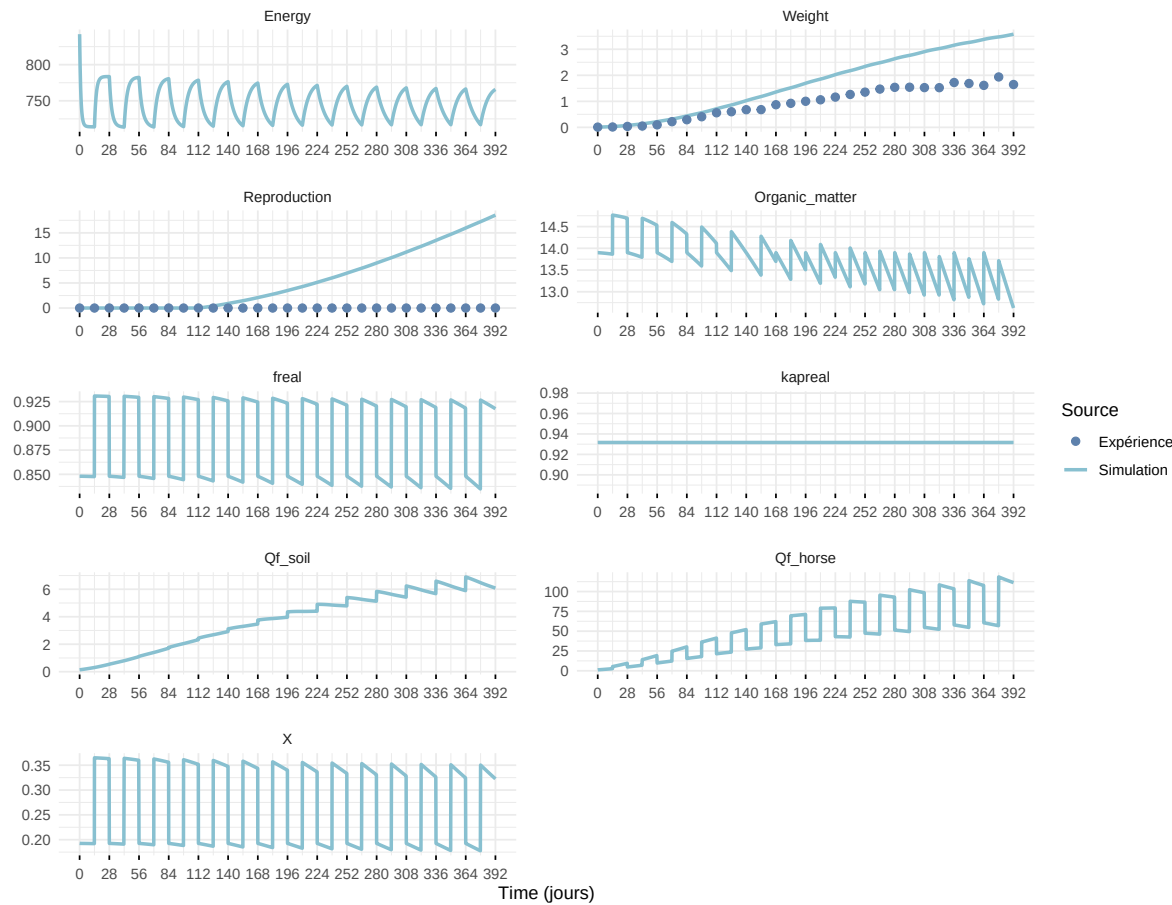
Expérience : Bart2019_XP2



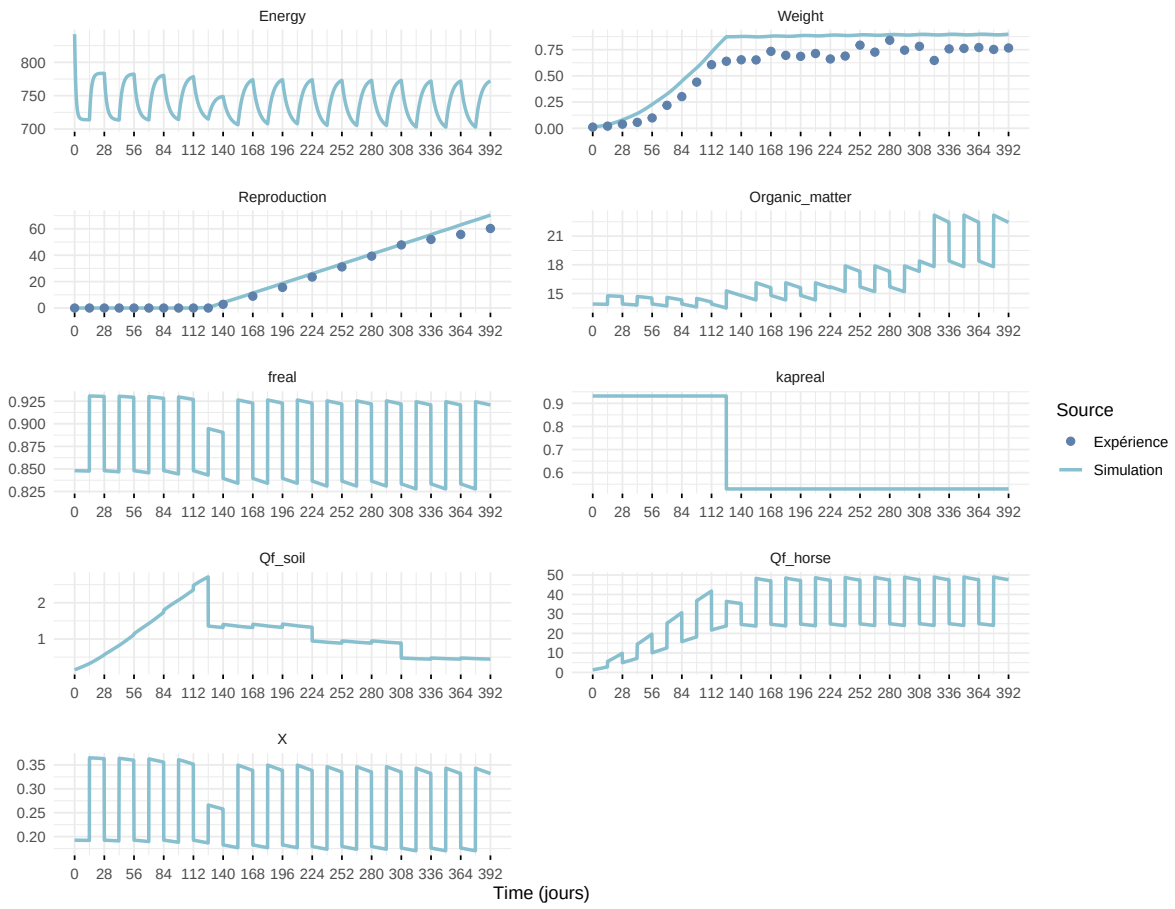
Expérience : Bart2019_XP3



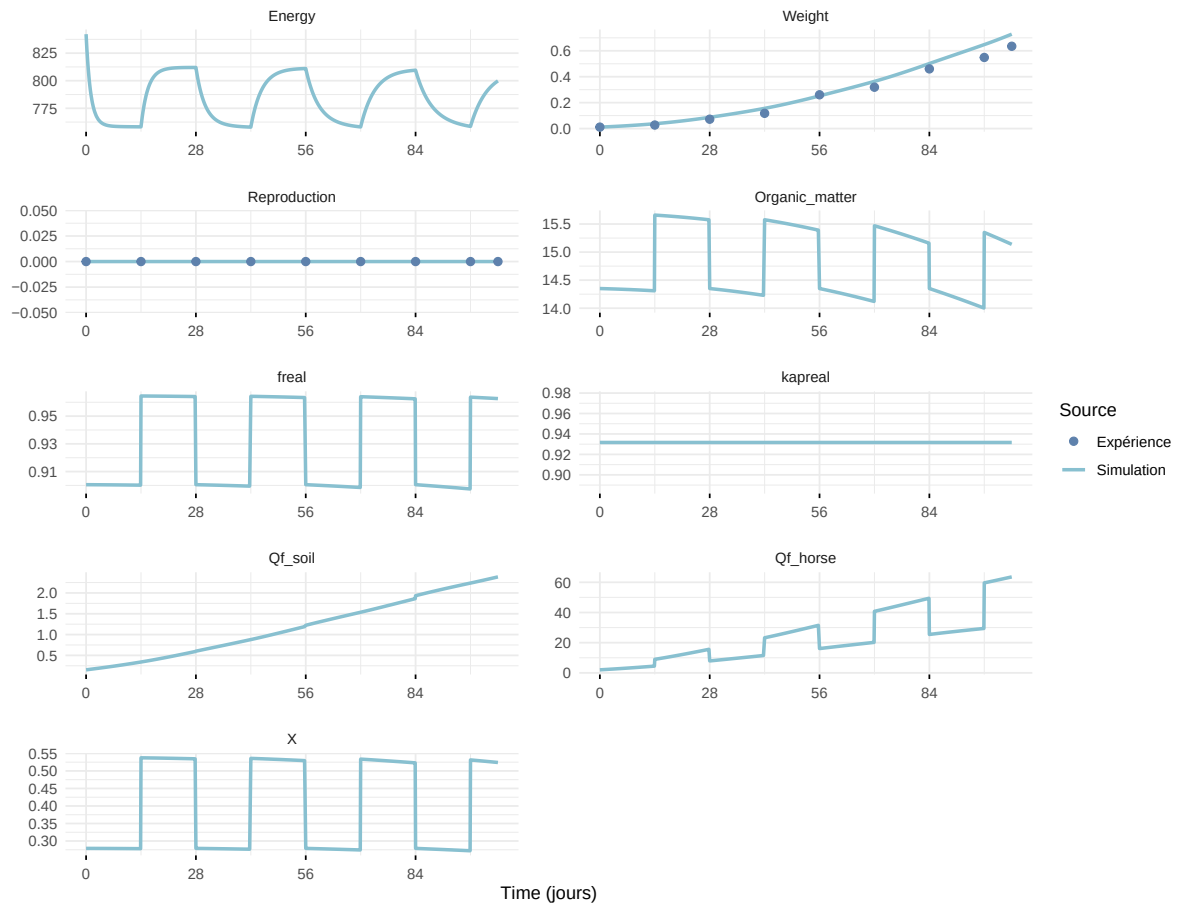
Expérience : Bart2019_XP4_1



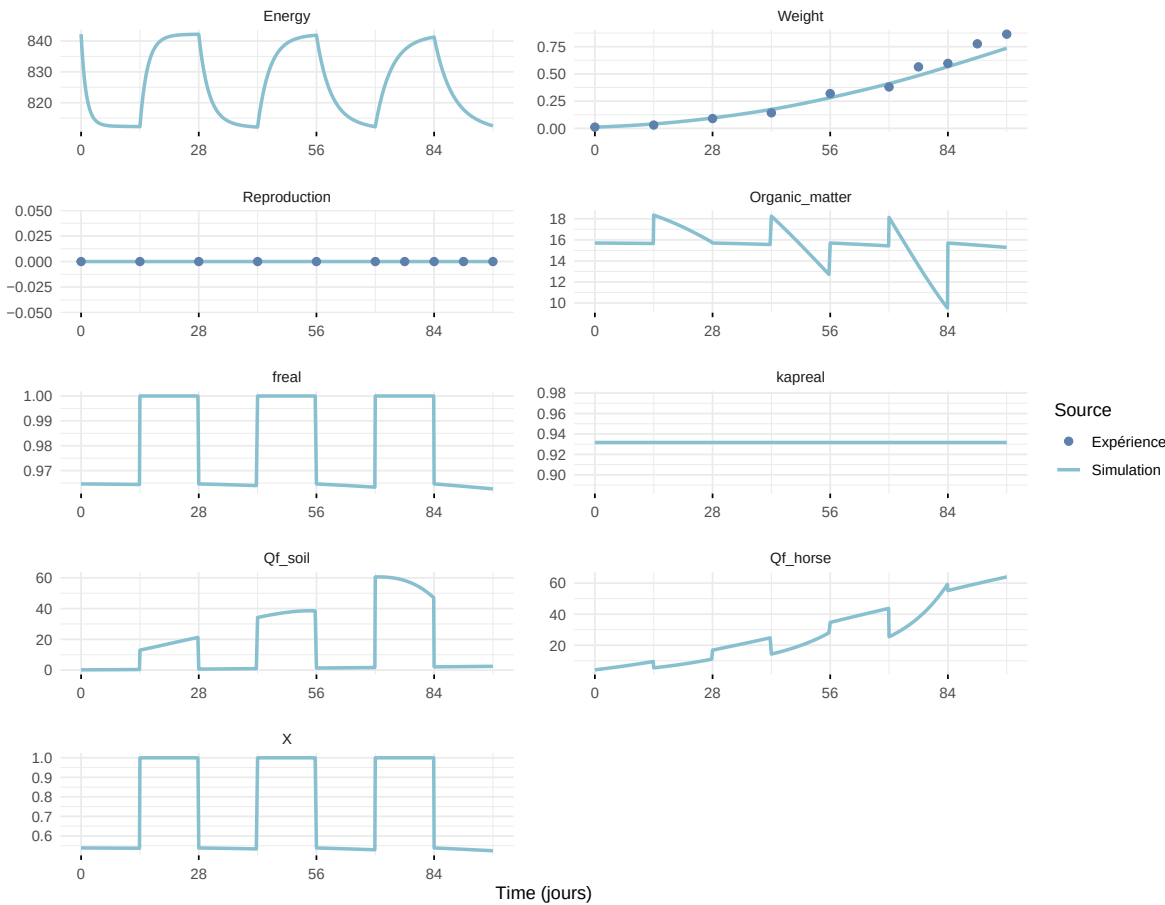
Expérience : Bart2019_XP4_2



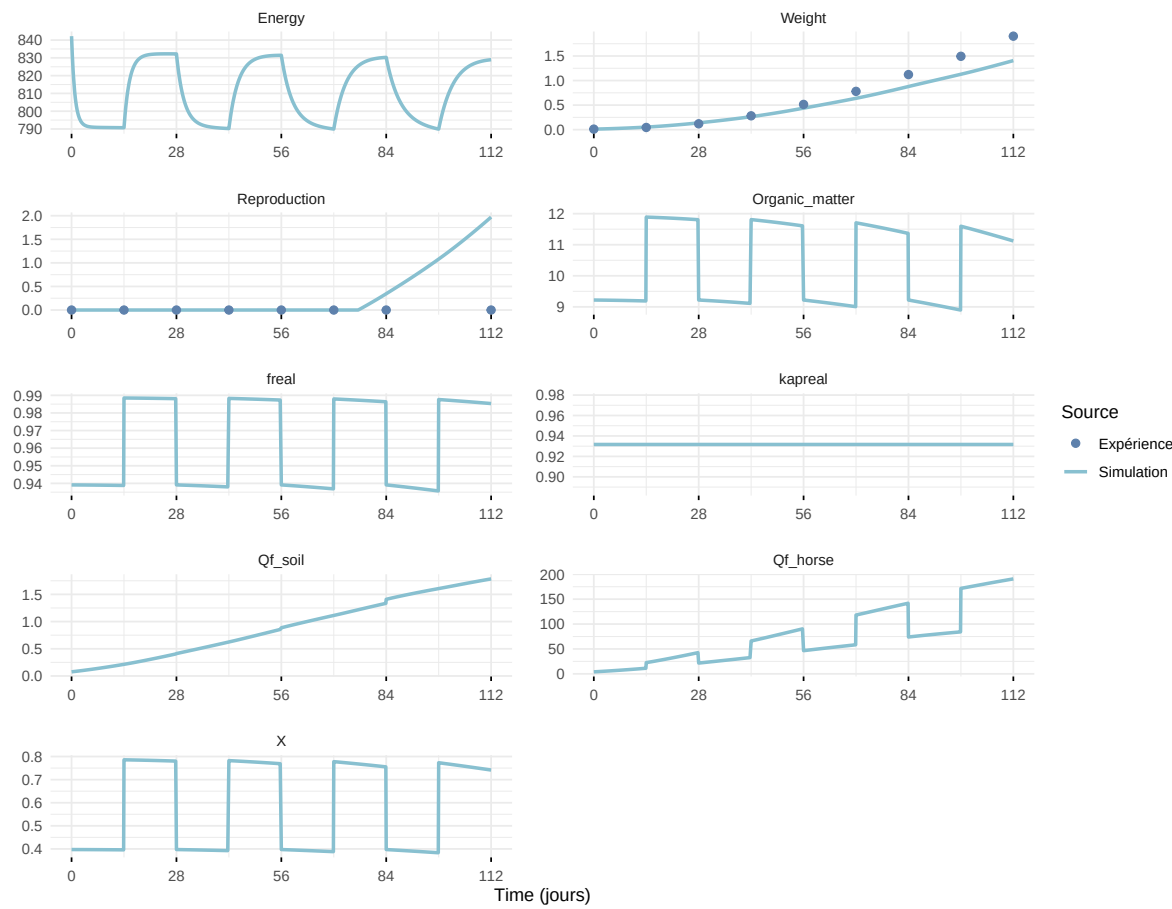
Expérience : Bart2019_XP5



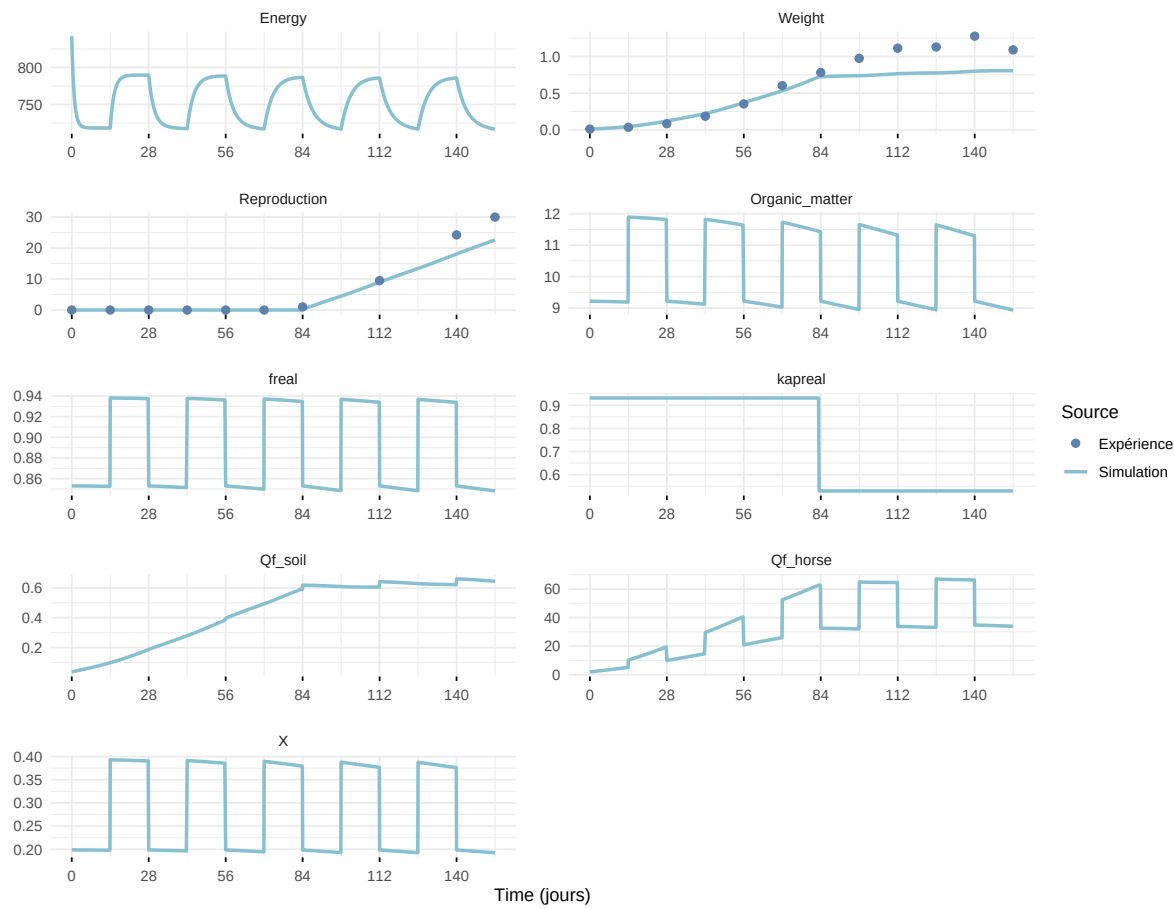
Expérience : Bart2020



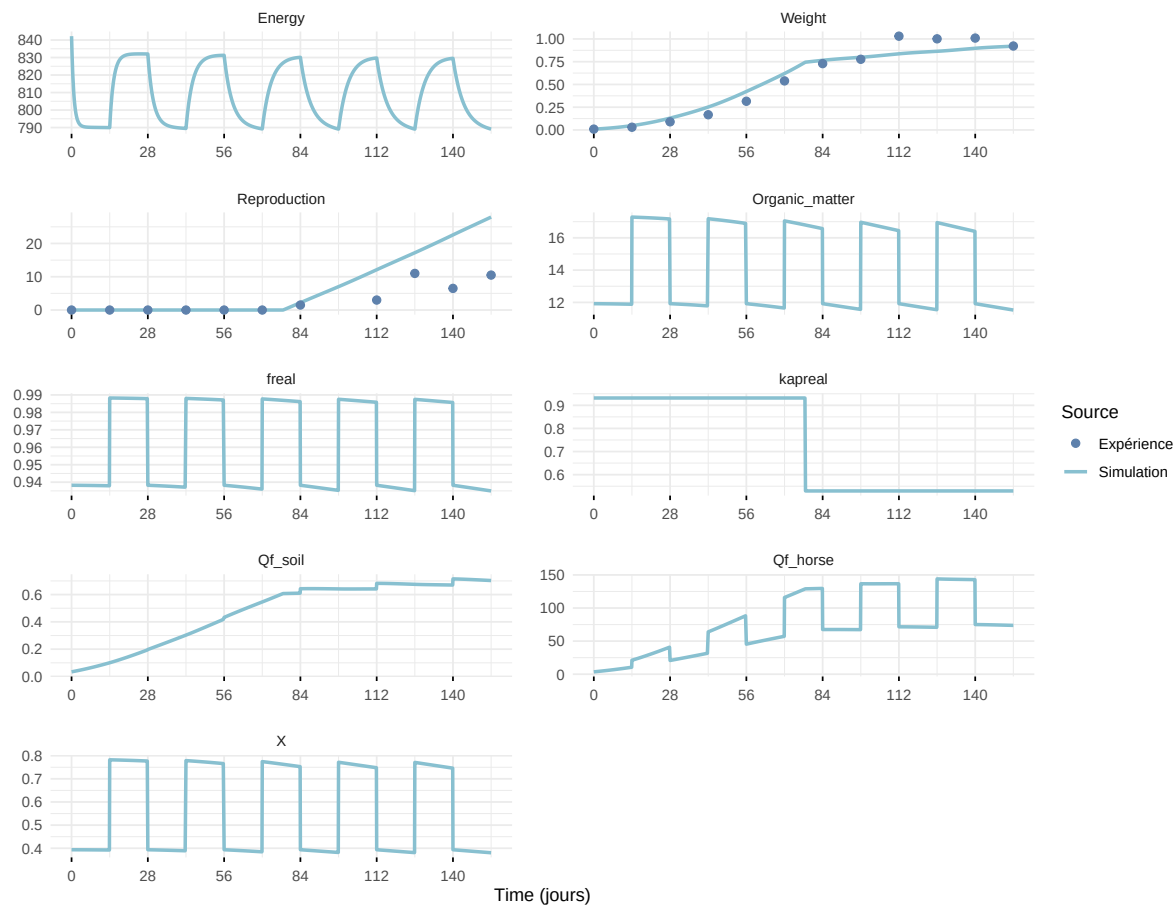
Expérience : Gollot2026_dens_D1



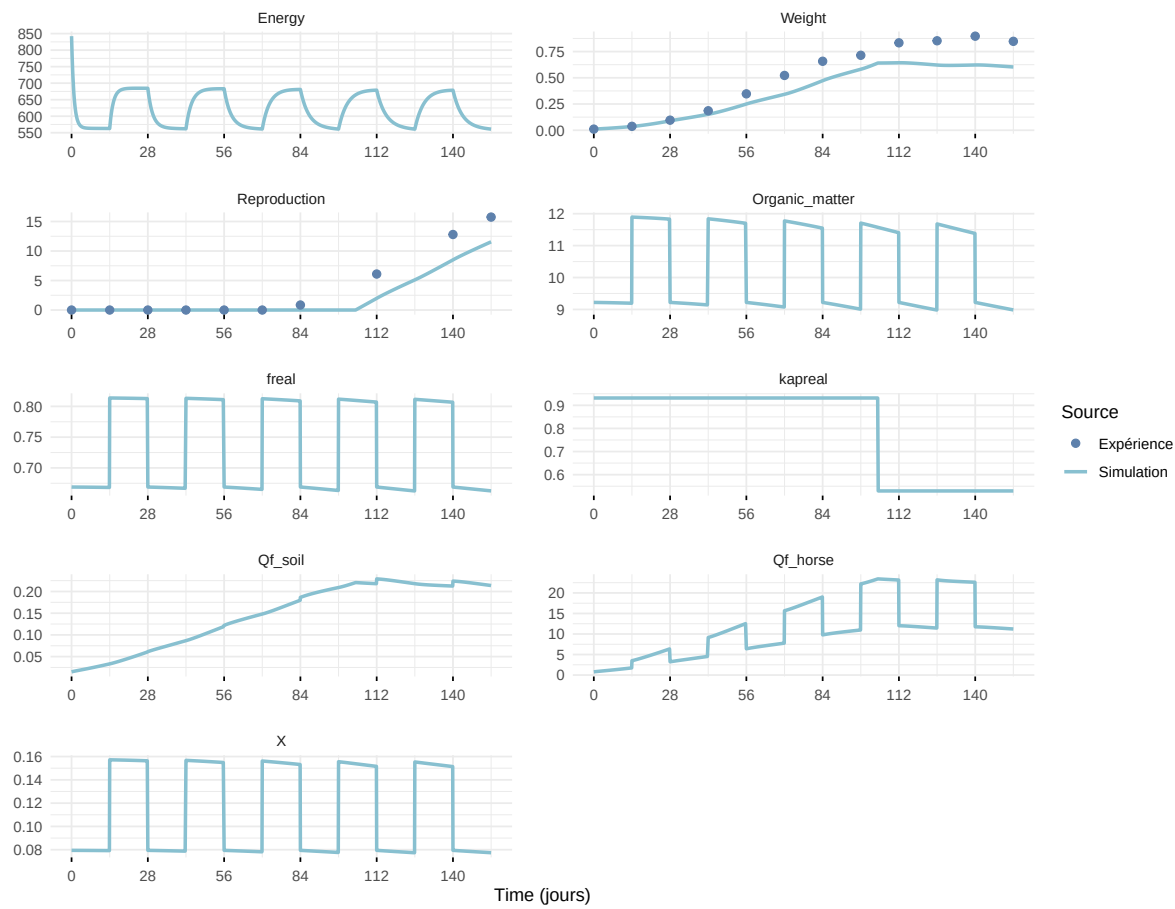
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

